CALEB BELLSMITH

BIOMEDICAL ENGINEERING STUDENT

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EDUCATION

University of Waterloo / Bachelor of Applied Science, Biomedical Engineering Co-op **Sept 2025 - Present**

- **WEEF Representative:** Nominated by class to evaluate technical proposals for student project funding and represented class interests.
- **Awards:** Governor General's Academic Medal, Sir Isaac Newton Physics Award, TDCH Valedictorian, HHCS Valedictorian.

TECHNICAL SKILLS

CAD/EDA: SolidWorks, Fusion360, AutoCAD, KiCad, Altium
Fabrication: CNC Machining, 3D Printing, Soldering, Manual Prototyping, Woodworking, Welding, Waterjetting, Laser Cutting
Programming: C++, C, Python, HTML, LaTeX, Visual Basic, MATLAB, CSS, JavaScript
Dev Tools: VSCode, CLion, Arduino IDE, Bamboo, GitHub, GitLab

EXPERIENCE

Automation Designer / Alchemy Nano **Sept 2026 - Present**
Part-Time

- Built a custom **in-line scanning system** for nanocomposite film quality testing from an array of **tiled global shutter cameras** and a custom **roll-to-roll converter**.
- Created the **vision software** end to end, covering **camera calibration**, focus and framing guidance, **defect detection**, and automated reporting with mapping of flagged regions.
- Optimized previously built **automated micro-graphing systems** and certified the custom system for **production-grade** use, including **electrical, mechanical, and safety certifications**.

Undergraduate Research Assistant / Wireless Sensors and Devices Laboratory **Sept 2026 - Present**
University of Waterloo / Research

- Designed and improved **autonomous assembly line robotics** learning systems, including **weight and ergonomic optimization** as well as **PCB design**.

Lab Automation Engineering Co-op / Alchemy Nano **Apr 2026 - Sept 2026**
Co-op

- Built an **automated micro-graphing robot** with a **MobileNetV3** frame-quality classifier (trained on **1000+** images, **99.3%** consistency), increasing lab efficiency by up to **440%** and accuracy by **40%**.
- Developed **Python** CV tools replacing manual optical measurement: **distortion analysis** via gradient edge detection, **polynomial baseline removal** and parallel batch processing, and **contact angle** measurement via **RANSAC** baseline fitting and a median of **3** geometric estimators.
- Designed a motorized film take-up roller with **quadrature encoder** closed-loop feedback, a standalone touchscreen interface, flash-stored resumable cycles, and a hardware **e-stop** built to Canadian electrical and workplace regulations.

Mechanical Designer / UWaterloo Space Soldering Team **2025 - Present**
Student Design Team

- Constructed a centrifuge for a **microgravity soldering experiment** selected for the **CAN-RGX Flight Campaign 2026** (NRC & CSA partnership). Technology is **patent pending** with commercial applications in space manufacturing and repair.
- Personally designed and fabricated the **wire feeding assembly**, **slip ring** exostructure, and core **rotation mechanism**, while running **FEA** and preliminary integration tests on all systems.

Electrical Designer / True North Bio-competition **2025 - 2026**
Student Design Team

- Designed a custom **PCB** as the complete electrical control framework for a **lower limb exoskeleton**, incorporating a **3-channel EMG** input to a **Teensy** for high-speed **ADC**, handed to an **ESP32** for **BLE** transmission, with **USB-C** power filtering and noise-suppression decoupling.

PROJECTS

Jolt Mouse / ESP32, IMU & EMG Sensors, BLE, Web Dev (Hack Canada 2025)

- Created a wearable **HMI** for amputee and upper-limb-disabled users, using **IMU** tilt to drive the cursor and **EMG** flexion to trigger clicks over a **BLE** pipeline, with a **full-stack web dashboard** to monitor and tune sensitivity and EMG positioning.

Precision Robotic Arm / ESP32, C++, Servo Control

- Designed and hand built a custom **5 DoF robotic arm** with a **1 N-m** grip, easy recharging, proportional control, and motion mirroring software.

Tilting Maze Platform / ESP32, Python, Computer Vision, PID

- Built a dual-axis servo platform commanded by a **Python** CV pipeline using **HSV** ball tracking, **Artificial Potential Field** navigation, **adaptive PID** auto-tuned every **60** frames with an **EMA** velocity predictor, and on-disk trajectory learning to optimize solve time.

Custom Cedar Strip Canoe / CAD, CNC Machining

- Computer modeled and fabricated a custom canoe using **CNC machined** internal molds and traditional woodworking.